the nucleus. Hence by adding and subtracting the mass of the Z electrons, we get

$$BE = \left(Zm_p + Zm_e + Nm_n - M - Zm_e\right)c^2 \quad (9.22)$$

$$BE = \left[Z\left(m_p + m_e\right) + Nm_n - M - Zm_e\right]c^2$$

where $m_p + m_e = m_H$ (mass of hydrogen atom)

$$BE = \left[Zm_H + Nm_n - (M + Zm_e)\right]c^2 \quad (9.23)$$

Here $M + Zm_e = M_A$ where M_A is the mass of the atom of an element ${}^{A}_{Z}X$.

Finally, the binding energy in terms of the atomic masses is given by

$$BE = \left[Zm_H + Nm_n - M_A\right]c^2 \quad (9.24)$$

Note

Using Einstein's mass-energy equivalence, the energy equivalent of one atomic mass unit

$1u = 1.66\times10^{-27}\times(3\times10^{8})^2$

$= 14.94\times10^{-11}\,J \approx 931\,MeV$

EXAMPLE 9.9

Compute the binding energy of ${}^{4}_{2}He$ nucleus using the following data: Atomic mass of Helium atom, $M_A\left(He\right) = 4.00260\,u$ and that of hydrogen atom, $m_H = 1.00785\,u$.

Solution:

Binding energy $BE = \left[Zm_H + Nm_n - M_A\right]c^2$

For helium nucleus, $Z = 2, N = A - Z = 4 - 2 = 2$

Mass defect

$$\Delta m = \left[\left(2\times1.00785\,u\right) + \left(2\times1.008665\,u\right) - 4.00260\,u\right] \quad \Delta m = 0.03043\,u$$

$$B.E = 0.03043u\times c^2$$

$$B.E = 0.03043\times931\,MeV = 28.33\,MeV$$

$$\left[\because 1uc^2 = 931\,MeV\right]$$

The binding energy of the ${}^{4}_{2}He$ nucleus is 28.33 MeV.

9.4.6 Binding energy curve

In the previous section, the origin of the binding energy is discussed. Now we can find the average binding energy per nucleon $\overline{BE}$. It is given by

$$\overline{BE} = \frac{\left[Zm_H + Nm_n - M_A\right]c^2}{A} \quad (9.25)$$

The average binding energy per nucleon is the energy required to separate single nucleon from the particular nucleus. When $\overline{BE}$ is plotted against A of all known nuclei. We get $\overline{BE}$ aveage curve as shown in Figure 9.24.

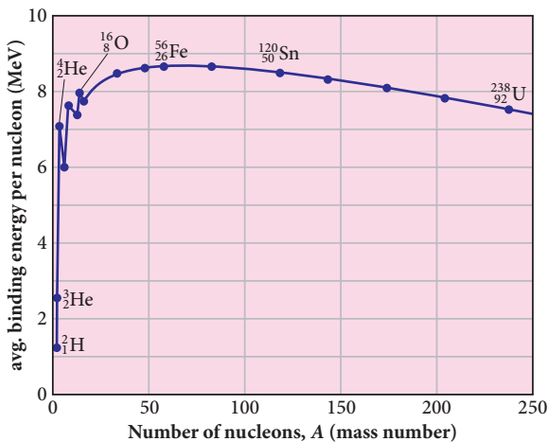


Figure 9.24 Avg. binding energy of the nucleons

Important inferences from of the average binding energy curve:

(1) The value of $\overline{BE}$ rises as the mass number increases until it reaches a maximum value of 8.8 MeV for A = 56 (iron) and then it slowly decreases.

(2) The average binding energy per nucleon is about 8.5 MeV for nuclei having mass number lying between A = 40 and 120. These elements are comparatively more stable and not radioactive.

UNIT-9(XII-Physics_Vol-2)_New.indd 167 21-07-2020 15:49:25